COMP 5300
Advanced Computer Architecture

Project 2

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1. Introduction

In the realm of computer architecture, the efficient organization and management of registers plays a pivotal role in the overall performance of a central processing unit (CPU). The register file is embedded within an 8-bit CPU. This register file comprises eight registers, each with an 8-bit width, denoted as Data 1, Data 2, and Data 3. In addition, there are three 3-bit wide address lines, identified as Address 1, Address 2, and Address 3. The implementation of this register file is realized through a combination of off-the-shelf integrated circuits, specifically the CY74FCT2574T, SN74HC151-Q1, M74HC238, and other essential logic gates.

The design and construction of this register file represent a fundamental aspect of computer architecture, where the efficient utilization of these integrated circuits plays a pivotal role in enabling the CPU to execute instructions with speed and precision. This introductory analysis will delve deeper into the intricate details of the register file's composition, elucidating how the CY74FCT2574T serves as an 8-bit register, the pivotal role of SN74HC151-Q1 in multiplexing, the M74HC238's function as a 3-to-8 decoder, and the incorporation of other crucial logic gates. Together, these components function harmoniously to facilitate the storage and retrieval of vital data within the CPU, thereby enabling the seamless execution of a wide array of instructions.

2. Implementation

Fig. 1 shows a register file inside an 8-bit CPU. It has 8 8-bit registers, that is, Data 1, Data 2 and Data 3 are all 8bit width, and Address 1, Address 2 and Address 3 are all 3-bit width. Implement this register file by using the following off-the-shelf chips.

(1) CY74FCT2574T: 8-bit Register
(2) SN74HC151-Q1: 8-to-1 Multiplexes
(3) M74HC238: 3-to-8 Decoder
(4) Other necessary gates that you need

For this implementation Logisim Eveloution version 3.8 will be used.

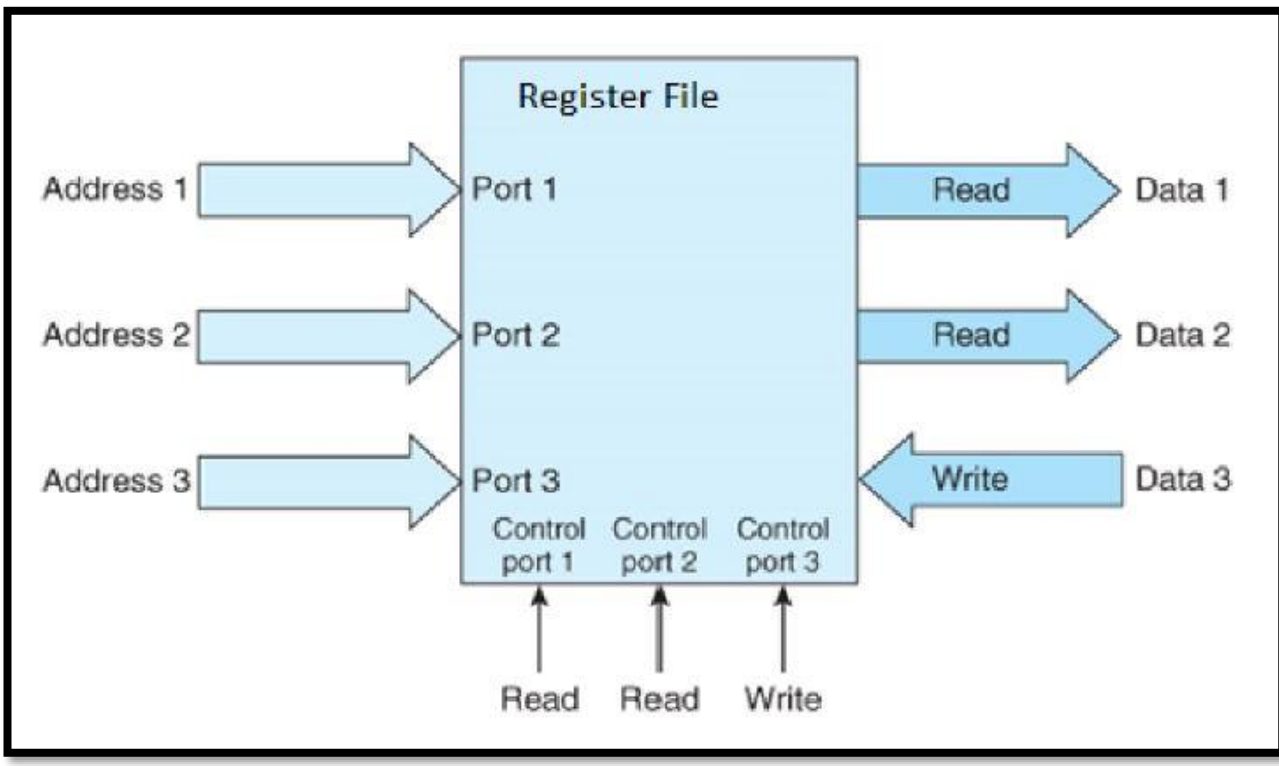

Figure 1. Register file to be implemented.

3. Building Logic Diagrams and Chip Representation

3.1 8-bit Register Logic Diagram

A register logic diagram depicts a digital circuit element used for storing binary data temporarily within a computer or digital system. It typically includes data input lines, clock signals, and control inputs for read and write operations. Registers are essential for tasks like data storage, temporary data manipulation, and buffering between different parts of a computer. They play a crucial role in data processing and movement within a CPU and are integral to the functioning of modern computing systems. The logic diagram Figure 1 represents the layout for the 8-bit register from the implementation sheet.

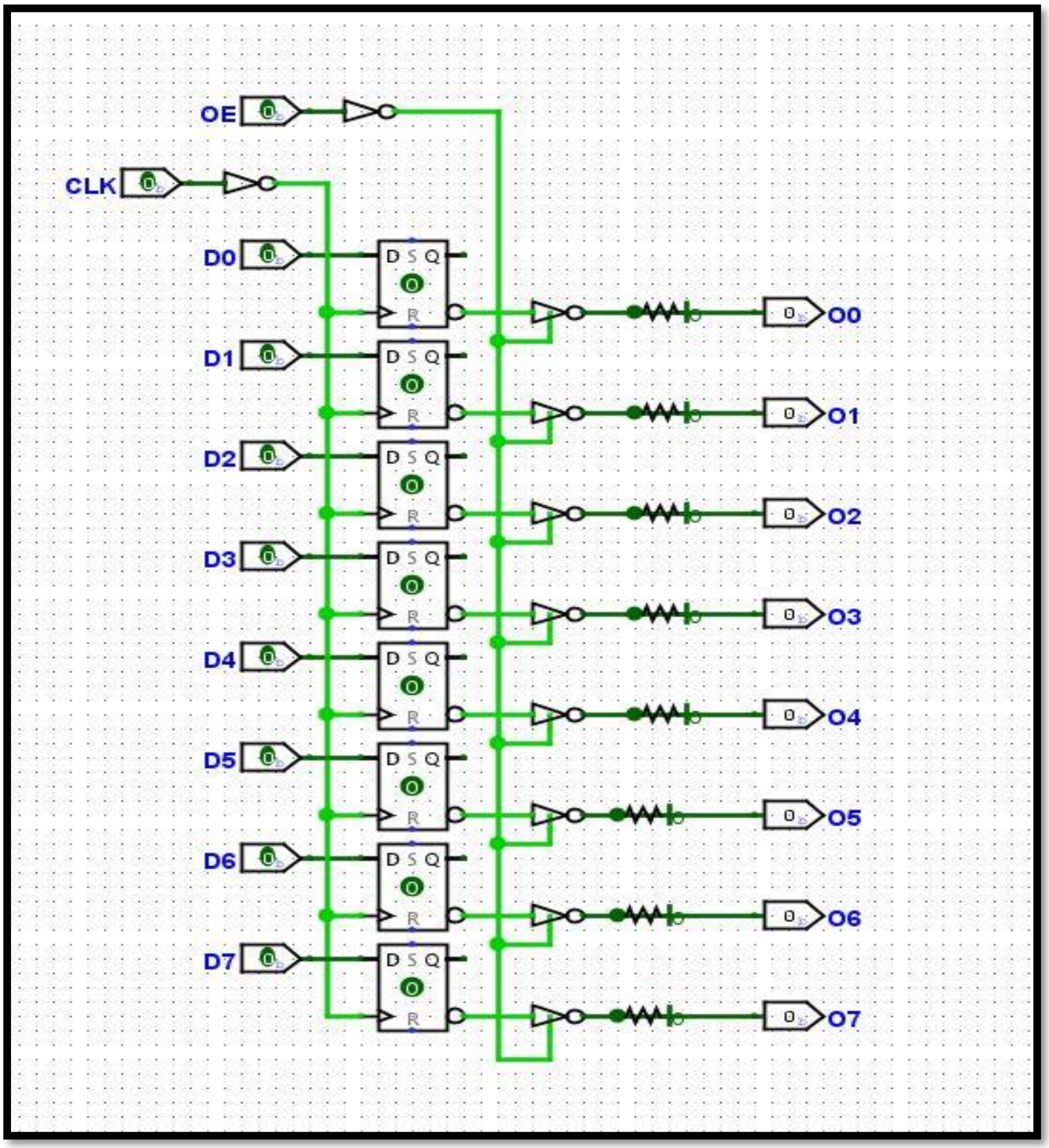

Figure 1. 8-bit Register Logic Diagram

3.2 8-bit Register Chip Representation

Figure 2 represents the 8-bit Register chip formatted from figure 1 8-bit Register Logic Diagram. The registers are used to store data in this circuit. Data is read from either the data 1 or data 2 pins depending on the control port selected. Each of the eight registers' first output pins is connected to one Multiplexer, and so on so that the output of one register is sent to the Multiplexer's output based on the selected lines. One multiplexer takes control port 1 as its enable and the other takes control port 2 as its enable. At data 1, the first multiplexer output is read, and at data 2, the second multiplexer output is read as well.

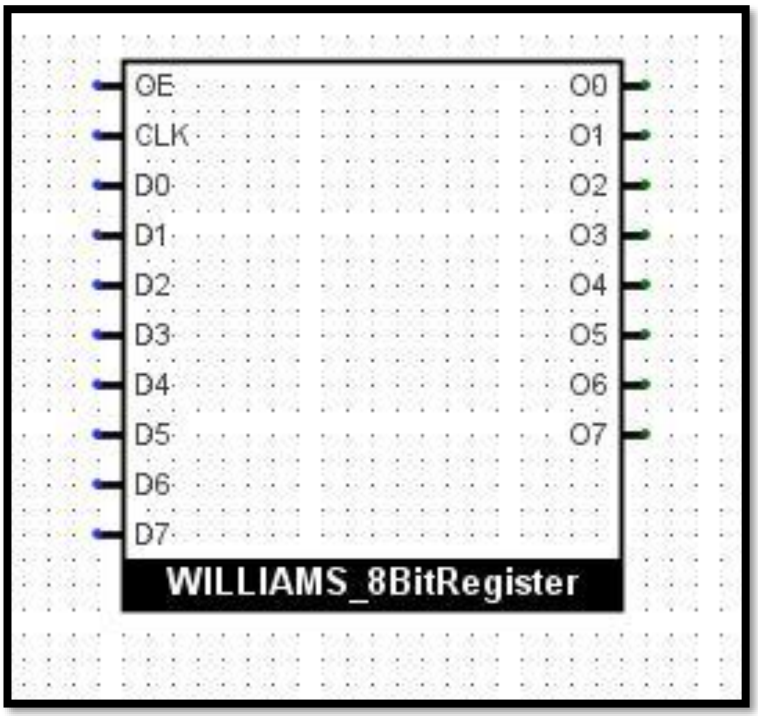

Figure 2. 8-bit Register Chip Representation

3.3 8 to 1 Multiplexer Logic Diagram

An 8-to-1 multiplexer is a versatile digital logic component commonly used in electronic circuits and digital systems. It serves as a data selector, allowing a single output line to be connected to one of eight input data lines based on the state of control inputs. This selective data routing is achieved by using three control lines that provide binary-encoded inputs, enabling the multiplexer to choose one of the eight data sources to be presented at the output. This functionality is invaluable in scenarios where different sources of data or signals need to be efficiently switched or routed to a single destination, making 8-to-1 multiplexers a fundamental building block in the design of digital circuits and systems. Figure 8 represents the 8 to 1 Multiplexer from the given data sheet.

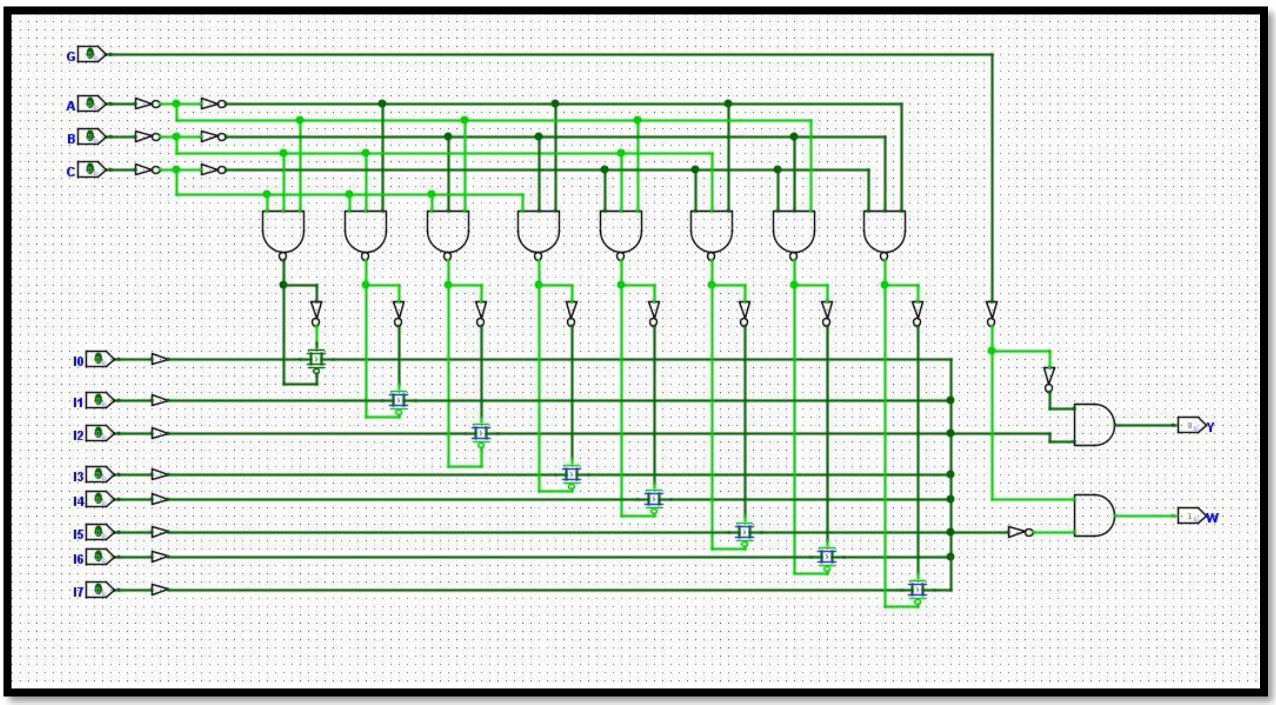

Figure 3. 8 to 1 Multiplexer Logic Diagram

3.4 8 to 1 Multiplexer Chip Representation

Figure 4 shows the 8 to 1 Multiplexer Chip Representation.

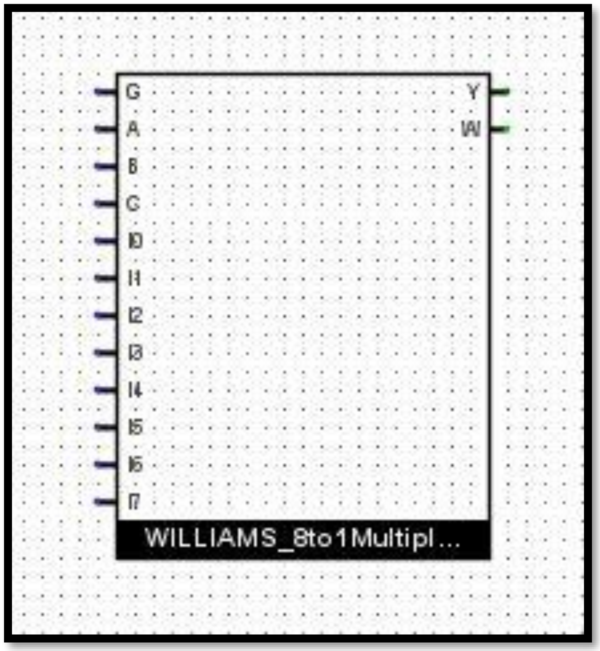

Figure 4. 8 to 1 Multiplexer Chip Representation

3.5 8 by 8 Multiplexer Logic Diagram

Figure 5 shows the connected 8 by 8 Multiplexer Logic Diagram.

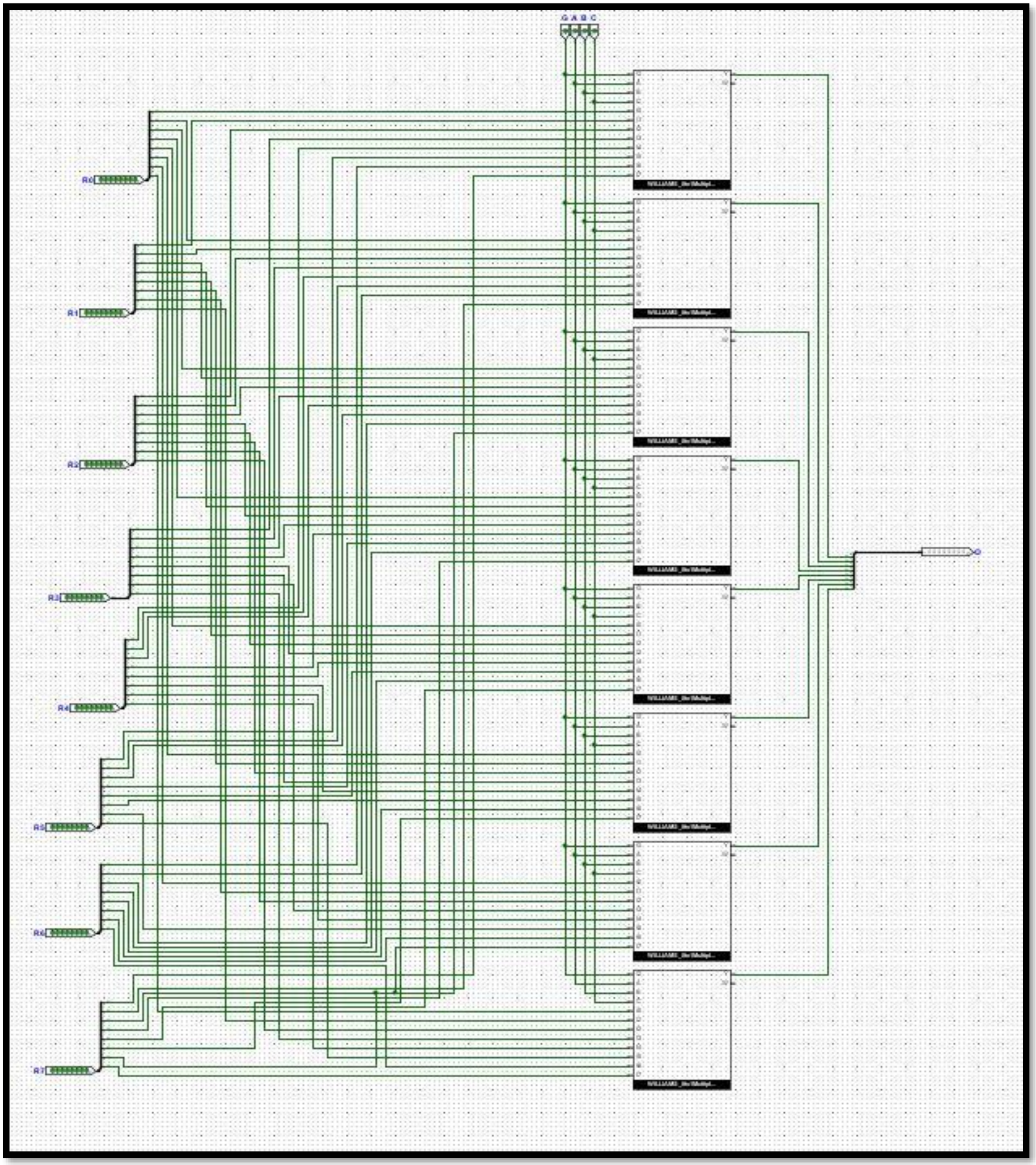

Figure 5. 8 by 8 Multiplexer Logic Diagram

3.6 8 by 8 Multiplexer Chip Representation

Figure 6 represents the 8-bit tristate buffer chip formatted from figure 5 8-bit Buffer Logic Diagram. The enable G1 pin serves as a Control Port for reading register data. The select lines of the multiplexer are width of 3-bits. This is accomplished by employing (2) 8-1 multiplexer pairs, with one pair using selection lines from Port 1 and the other using selection lines from Port 2

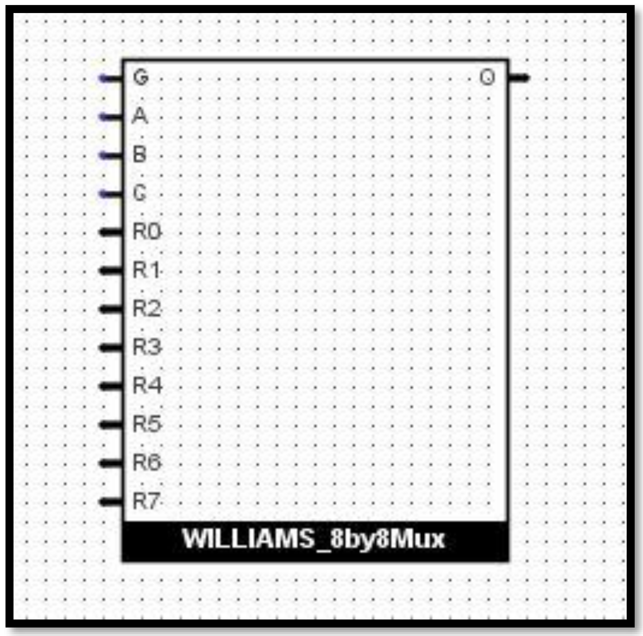

Figure 6. 8 by 8 Multiplexer Chip Representation

3.7 3 to 8 Decoder Logic Diagram

Figure 7 represents the 3 to 8 Decoder Logic Diagram. A 3-to-8 decoder is a crucial digital logic component commonly found in digital systems and computer architecture. It is designed to take a 3-bit binary input and use it to select one of eight output lines, which can be individually activated. Essentially, the 3-to-8 decoder serves as a data distributor, with each output line corresponding to a unique binary combination of the three input bits. When a specific binary input is applied, only the corresponding output line is activated, while all others remain inactive. This functionality is particularly useful for tasks such as memory address decoding, control signal generation, and data routing in various digital applications, making 3-to-8 decoders a fundamental building block for implementing complex digital systems and logic circuits.

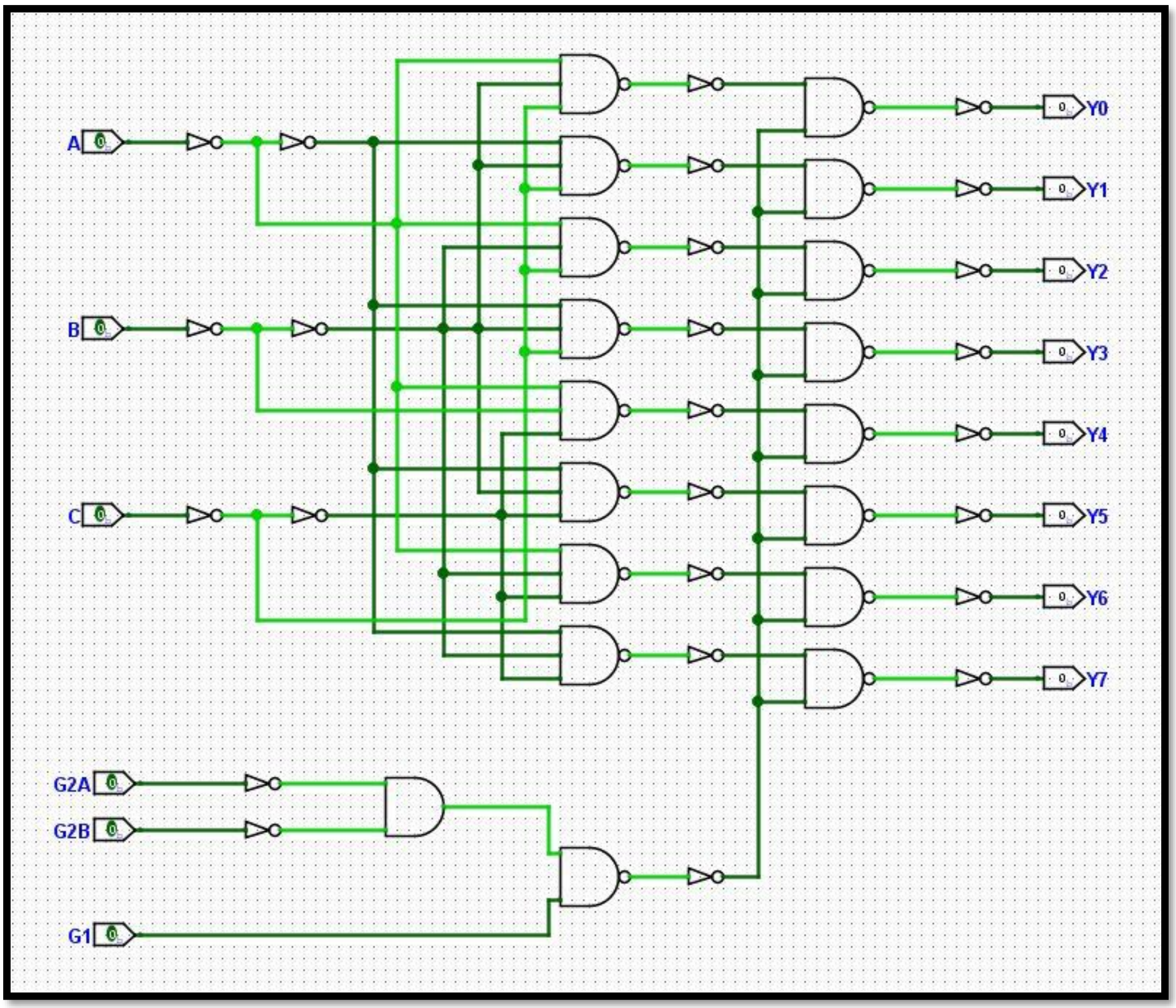

Figure 7. 3 to 8 Decoder Logic Diagram

3.8 3 to 8 Decoder Chip Representation

Figure 8 represents the 3 to 8 Decoder Chip Representation created from Fig 7 logic diagram. The control port 3 pin and the inverted decoder outputs are combined using 8 AND gates. When the control port 3 pin is LOW no write operation is happening, The And gates are used to enable the registers. Data 3 can WRITE data when the control port 3 is HIGH and when the output is low on one of the decoders. Based on the decoder's selection lines, this circuit enables a decoder to enable either of the registers. The decoder has 8 outputs that are connected to the enable lines of the 8 registers. The output lines decide which register is enabled, and data 3 can write data to that register. The decoder's enable lines G2A, G2B, and G1 are all connected to Control Port 3, and that line is used to turn it on.

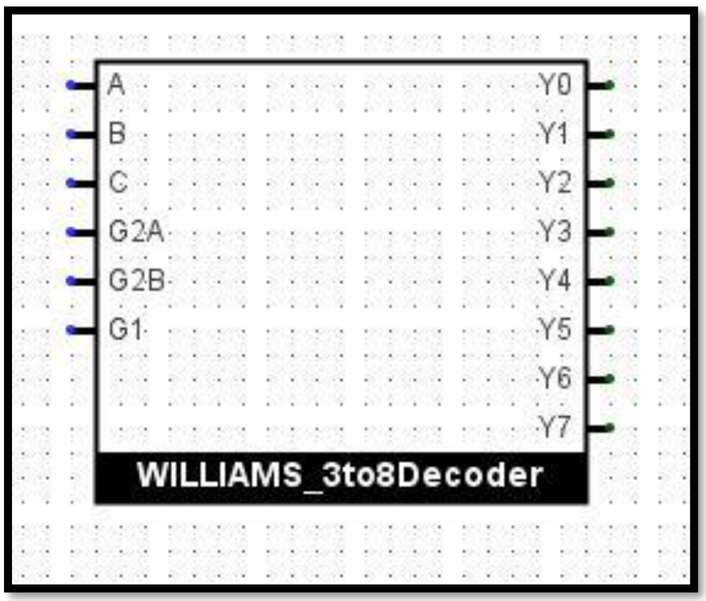

Figure 8. 3 to 8 Decoder Chip Representation

3.9 8-bit Buffer Logic Diagram

A buffer logic diagram illustrates a simple electronic circuit designed to transmit digital signals from one point to another without altering their content or timing. It typically consists of an input, an output, and a high-impedance state controlled by an enable signal. When enabled, the buffer allows data to flow from input to output. The logic diagram Figure 9 represents the layout for the 8-bit tristate buffer from Texas Instrument.

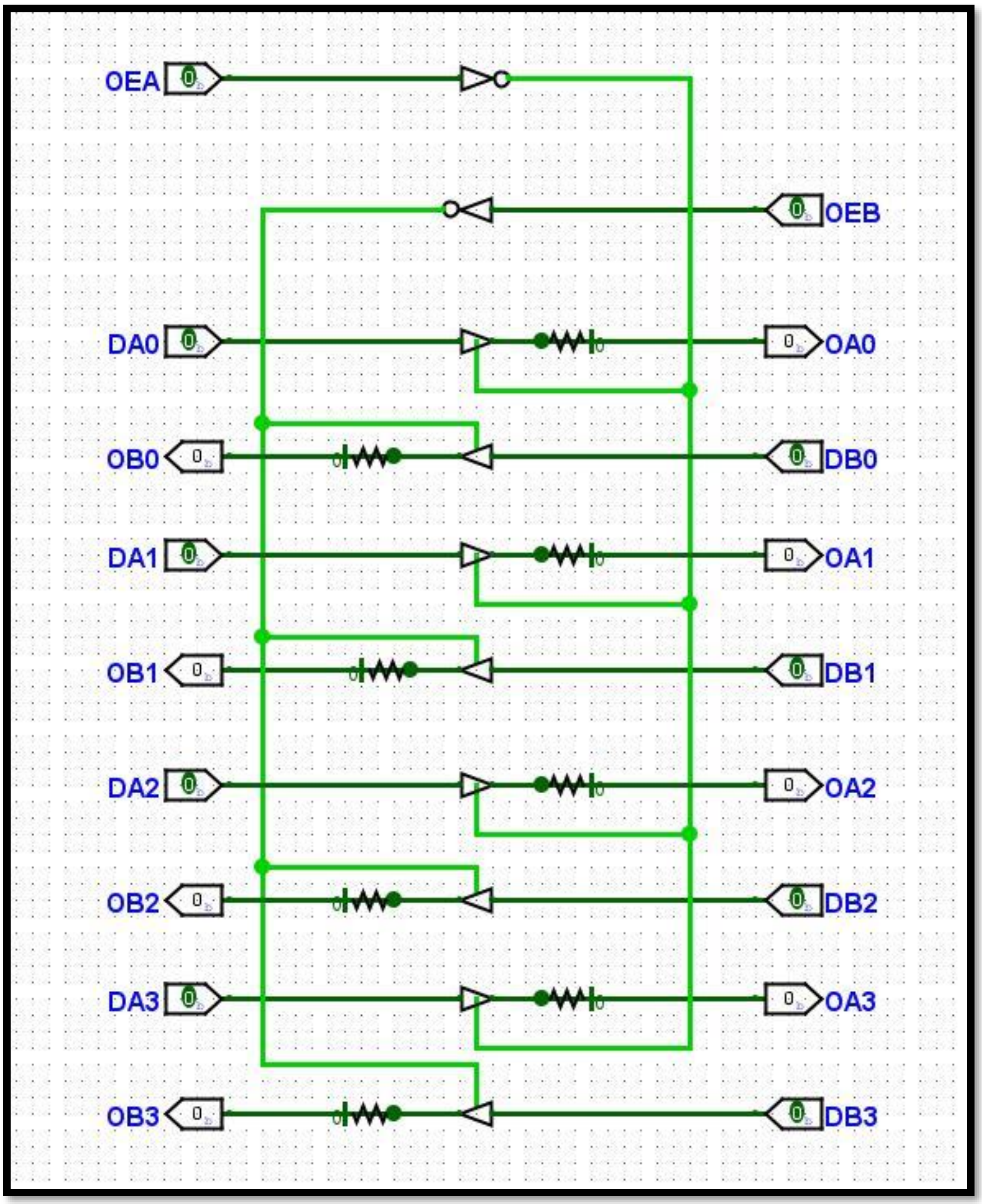

Figure 9. 8-bit Buffer Logic Diagram

3.10 8-bit Buffer Chip Representation

Figure 10 represents the 8-bit tristate buffer chip formatted from figure 9 Logic Diagram. The Buffer is used to prevent the Data 3 input from being WRITE to all the Register File's registers. The buffer is controlled by port 3 line. When the pin is LOW, the buffer stops WRITE data from the Data 3 input to the registers. The buffer allows the Data 3 input to WRITE data to the register when the control port 3 pin is high.

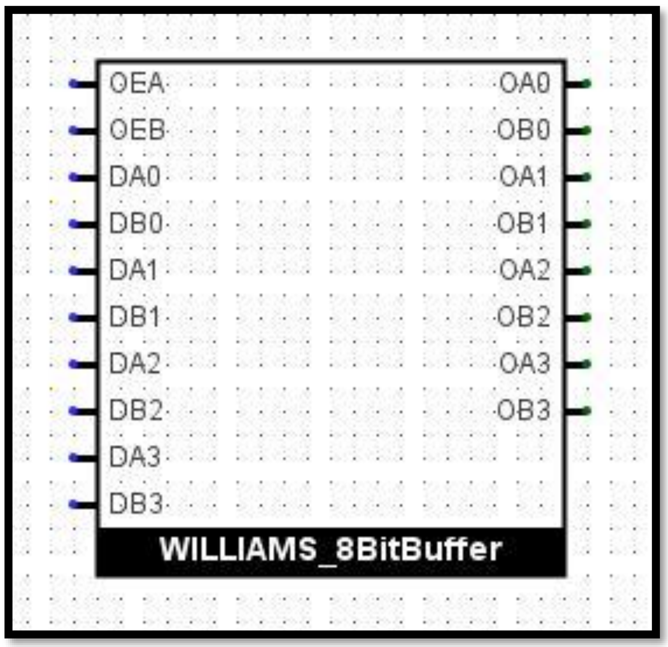

Figure 10. 8-bit Buffer Chip Representation

3.11 Implemented Register File Logic Diagram

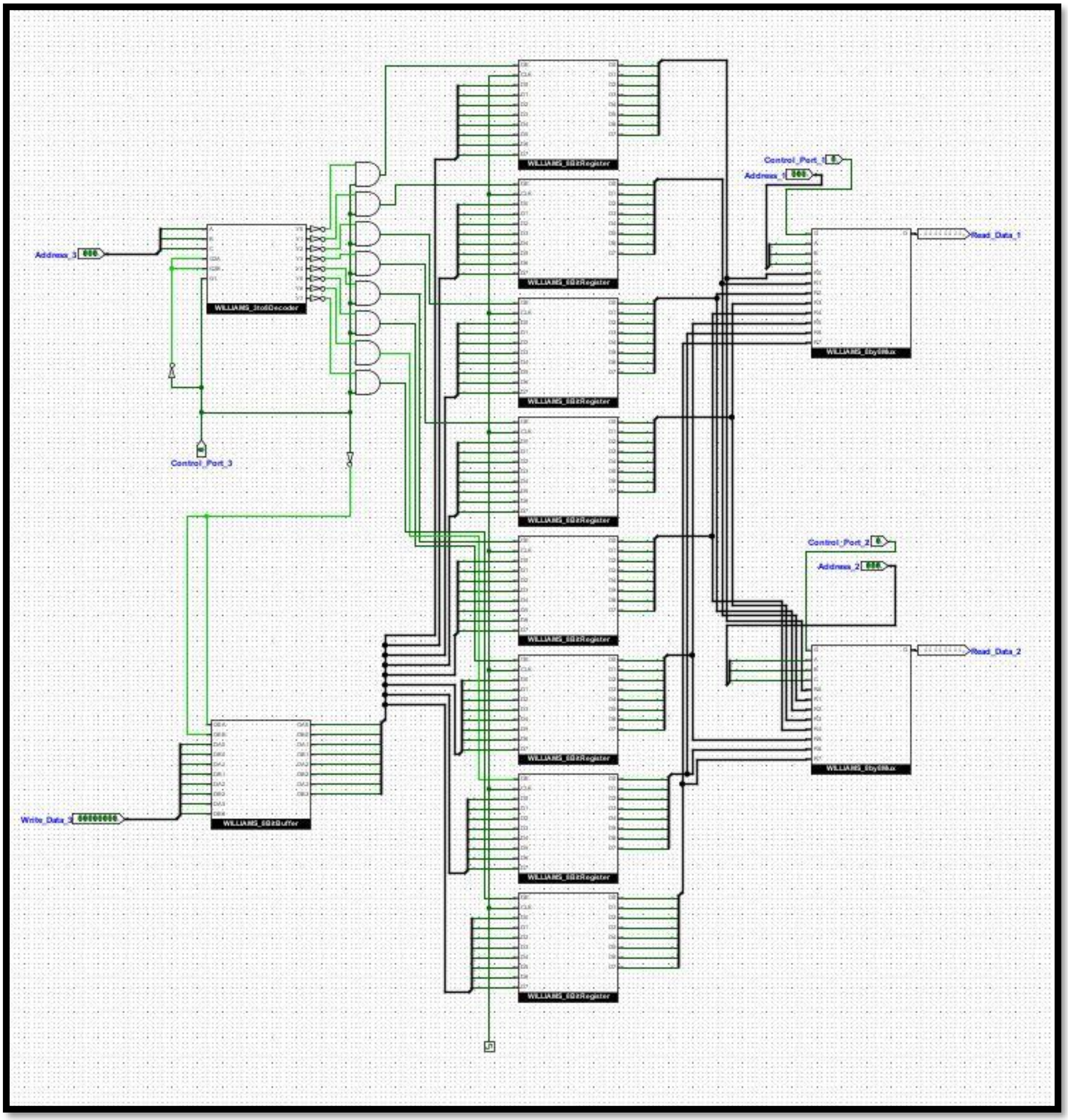

Fig 11. Implemented Register File Logic Diagram

4. Conclusion

4.1 Implemented Register File Diagram

A register file is a vital component within a computer's central processing unit (CPU), responsible for the storage and management of essential data. In this register file configuration, three 3-bit wide address lines, labeled as Address 1, Address 2, and Address 3, are employed to select specific registers within the file. Each register in this register file is 8 bits wide, which means it can store 8 bits of data. Two read data ports, labeled as Data 1 and Data 2, enable the retrieval of information from the specified registers based on the selected addresses. Additionally, there is one write data port, Data 3, allowing for the modification and storage of data in the selected register.

To control the operations of this register file, three control ports, designated as Control 1, Control 2, and Control 3, are utilized. These control ports play a pivotal role in specifying whether a read or write operation is to be performed, determining which register is being accessed, and facilitating data transfer between the CPU and the register file. This organization and interplay of addresses, data ports, and control ports are essential for the seamless functioning of a CPU, enabling it to efficiently manage and manipulate data during the execution of various instructions, ultimately contributing to the overall computational capabilities of the system.

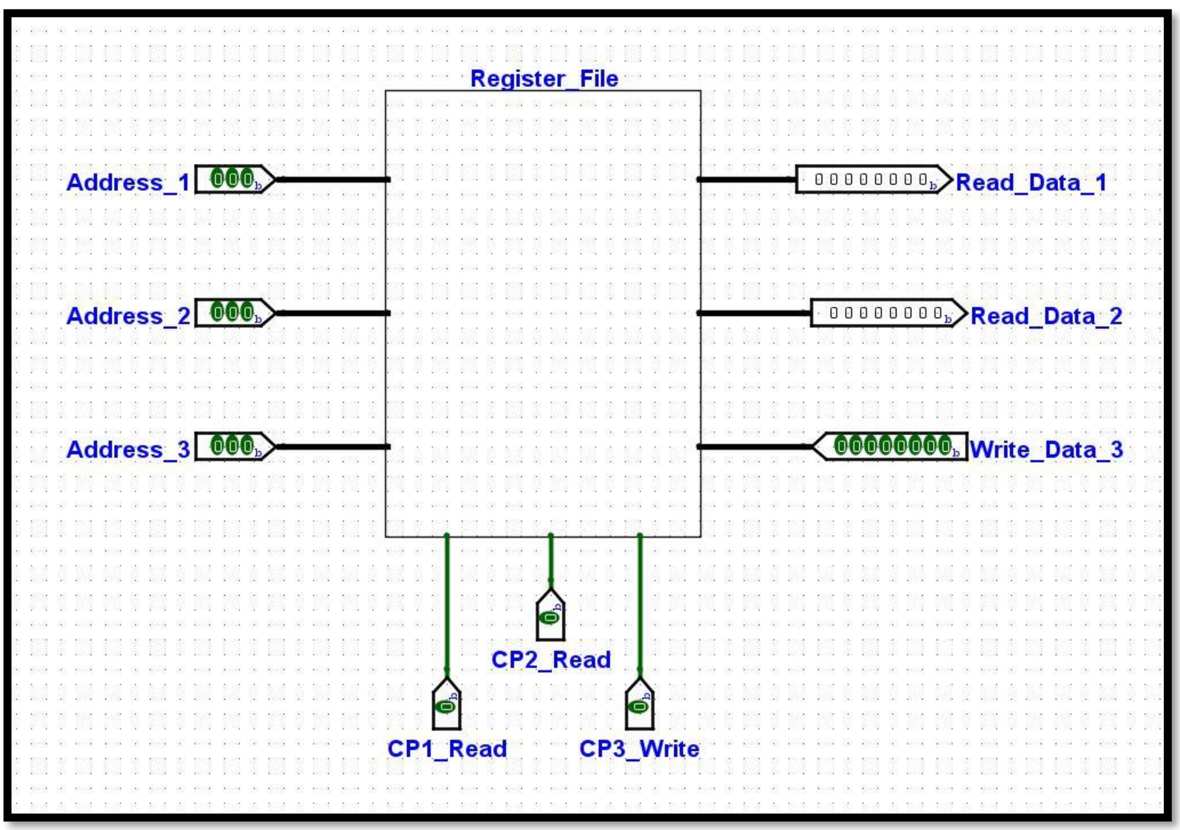

Figure 12. Implemented Register File Diagram

4.2 Implementation Description

Data can be read from either the data 1 or data 2 lines depending on the right control port selection. The 8 registers are used to store data in this circuit. The data for which registers should be read at the data 1 line is determined by port 1, which is used to enable read. Similarly, data 2 line of control port 2 is used to select which registers should have their data read from. Writing data from data 3 is made possible by control port 3, and port 3 decides which register the data should be written to. The 3-bit wide addresses 1, 2, and 3 are connected at Ports 1, 2, and 3. The addresses of data 1 and data 2 serve as the basis for reading the data stored in the registers. Write data to the registers using data 3, is determined by address 3.